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Topological exciton insulator

Zhongqi Xiu

2026.09.10

Device structure

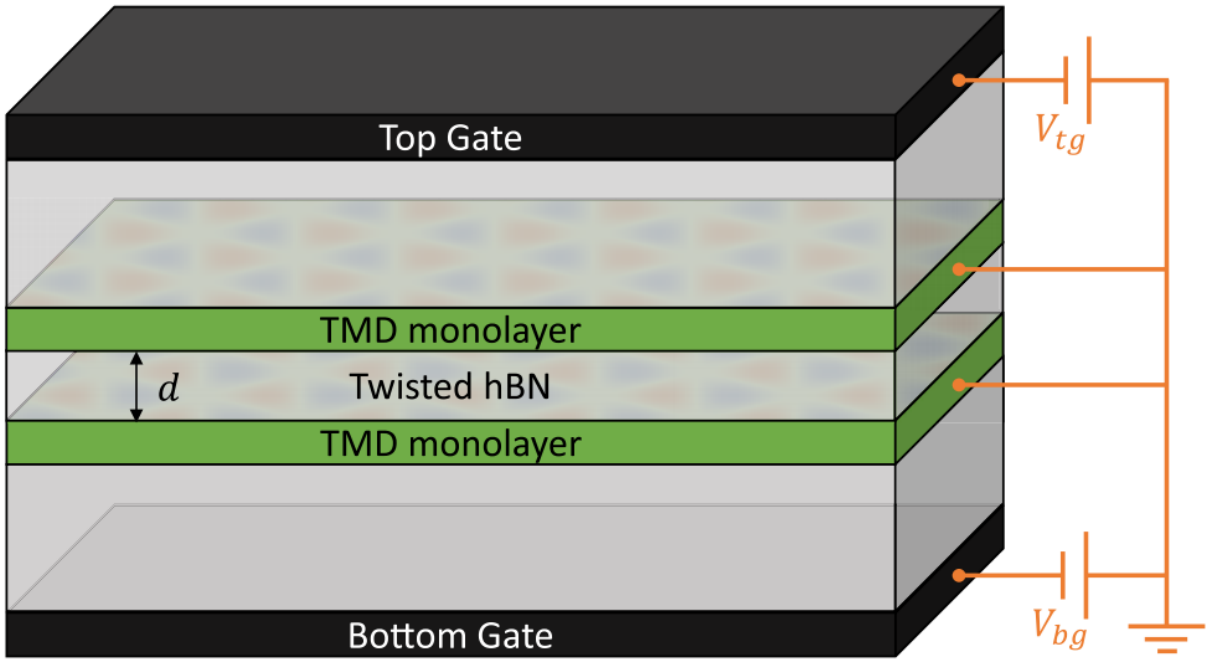
Topological Excitonic Insulators in Electron Bilayers Modulated by Twisted Hexagonal Boron Nitride

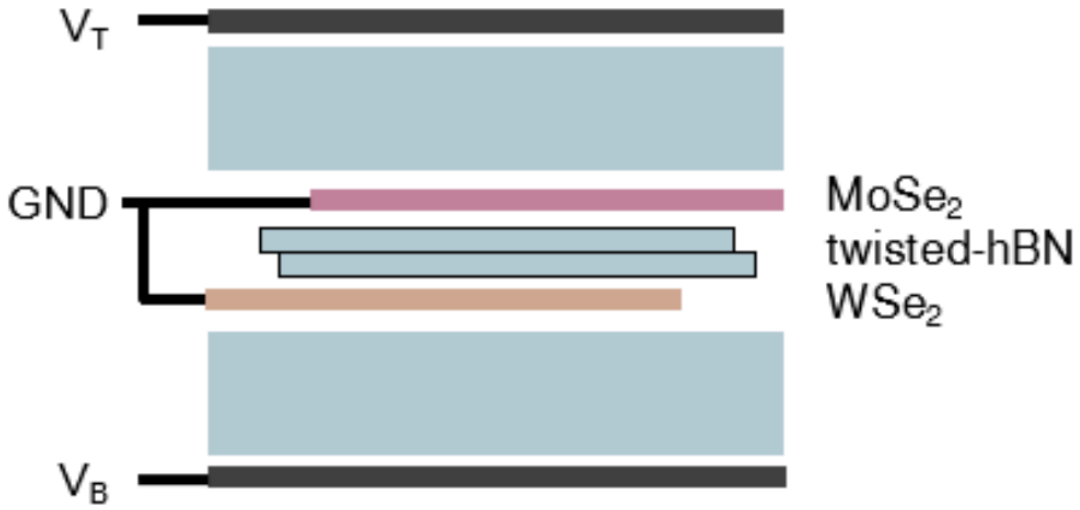
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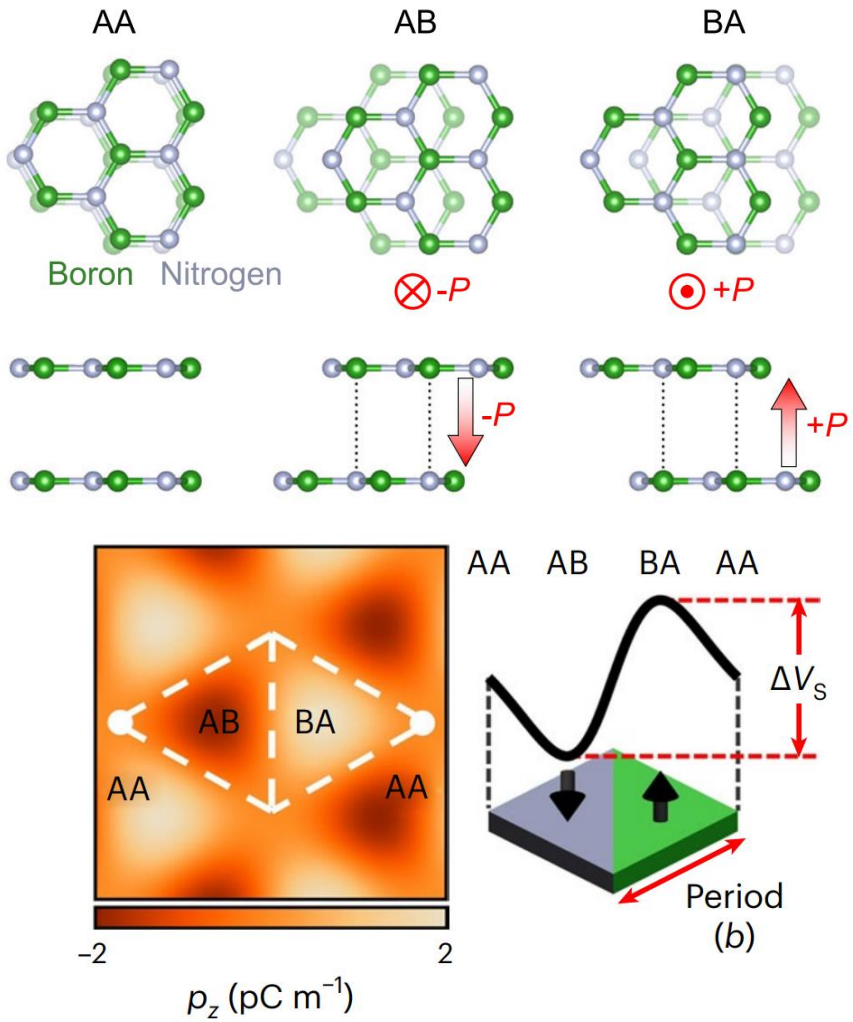


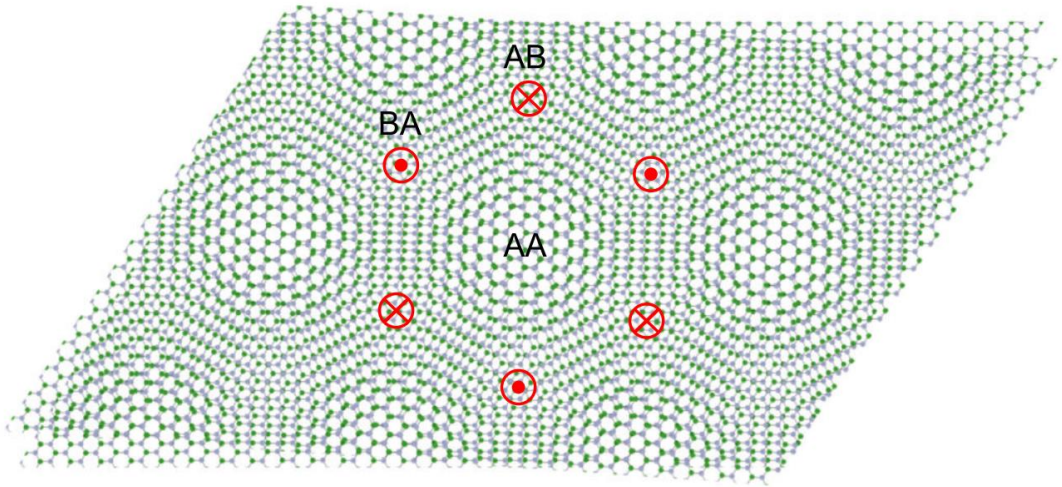


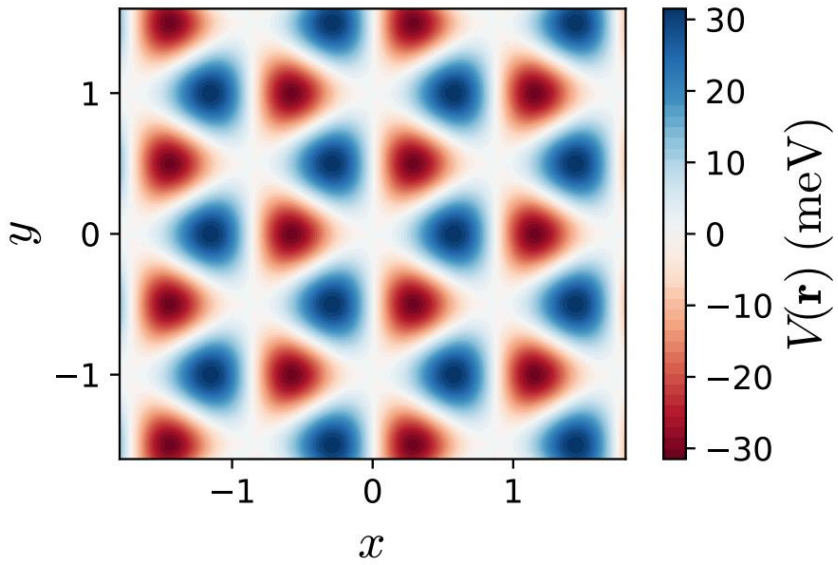
arXiv:2509.13159

Moiré polar substrate

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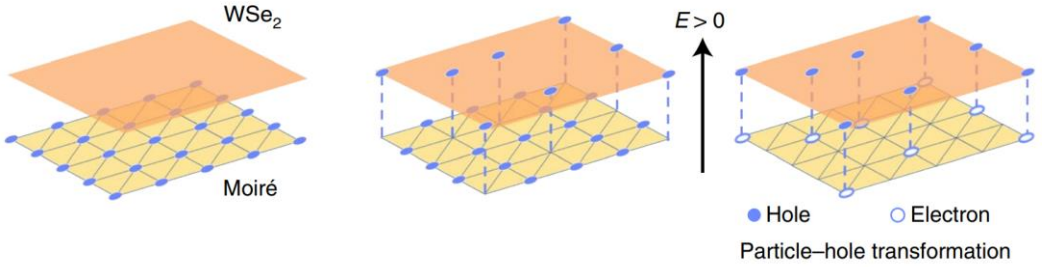


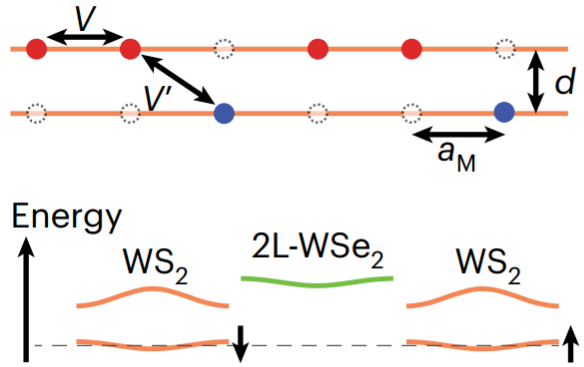


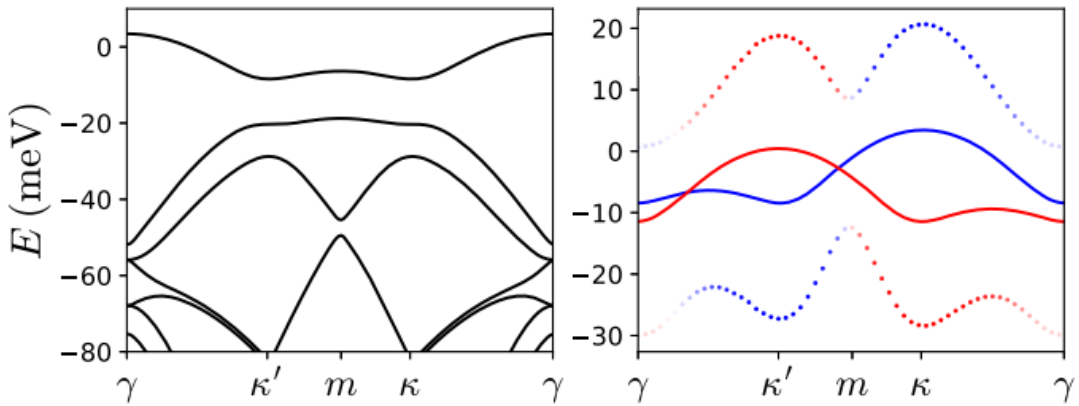
Nat. Mater. 23, 65–70 (2024)

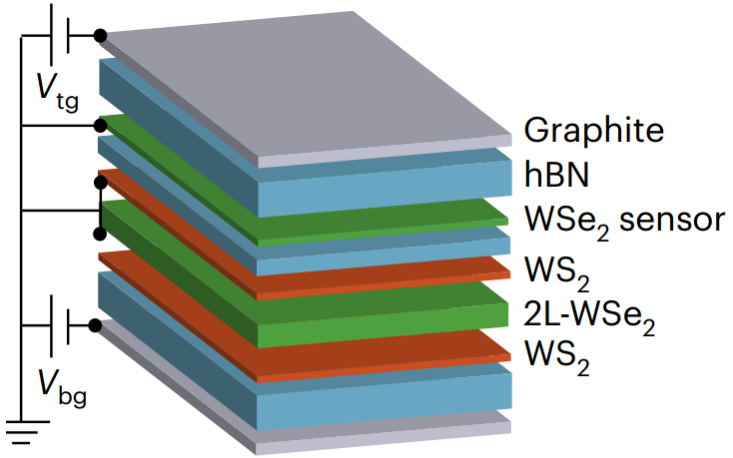
Band structure

The exciton is a transfer between two valence minibands



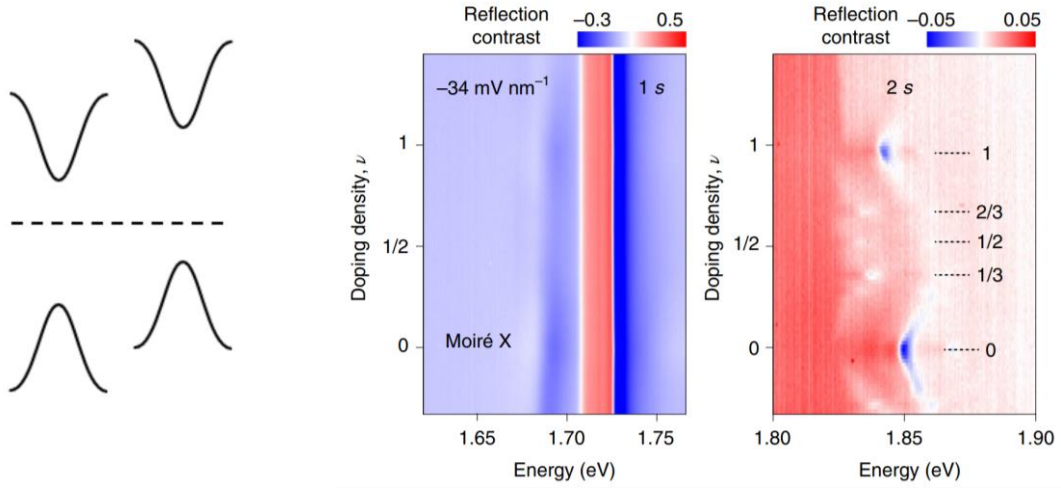


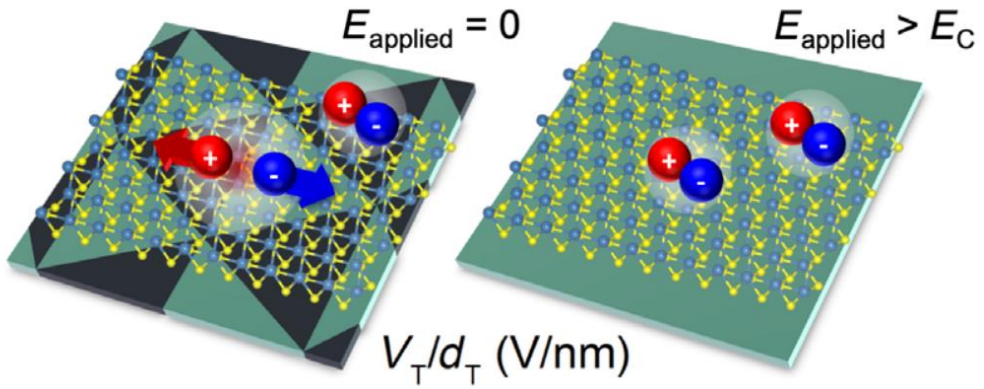


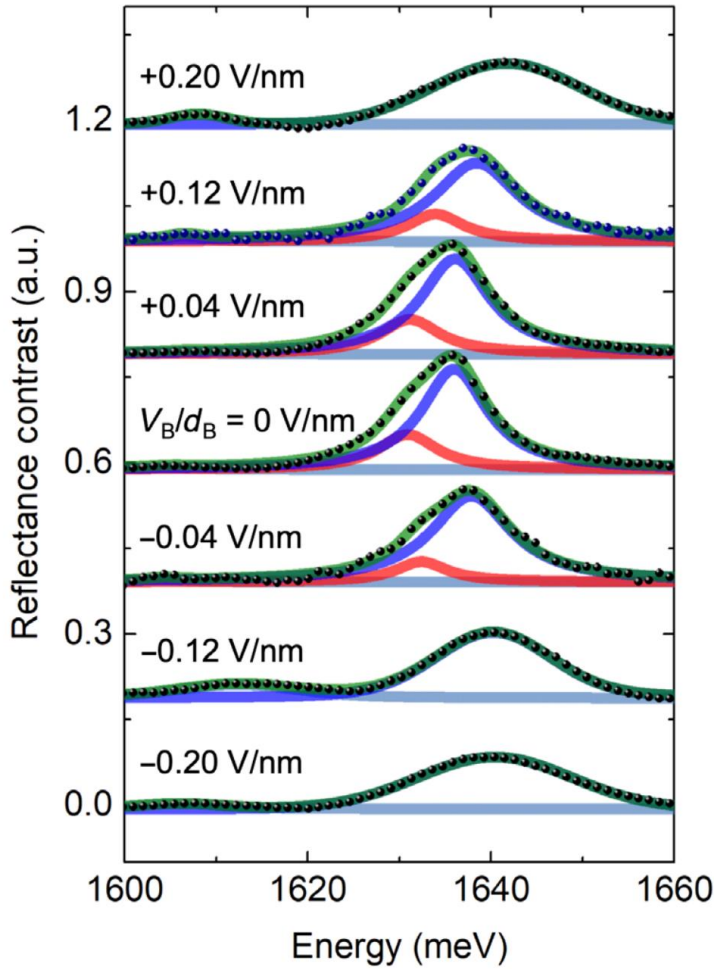


Nat. Phys. 18, 395–400 (2022)

E field modulate Vm?



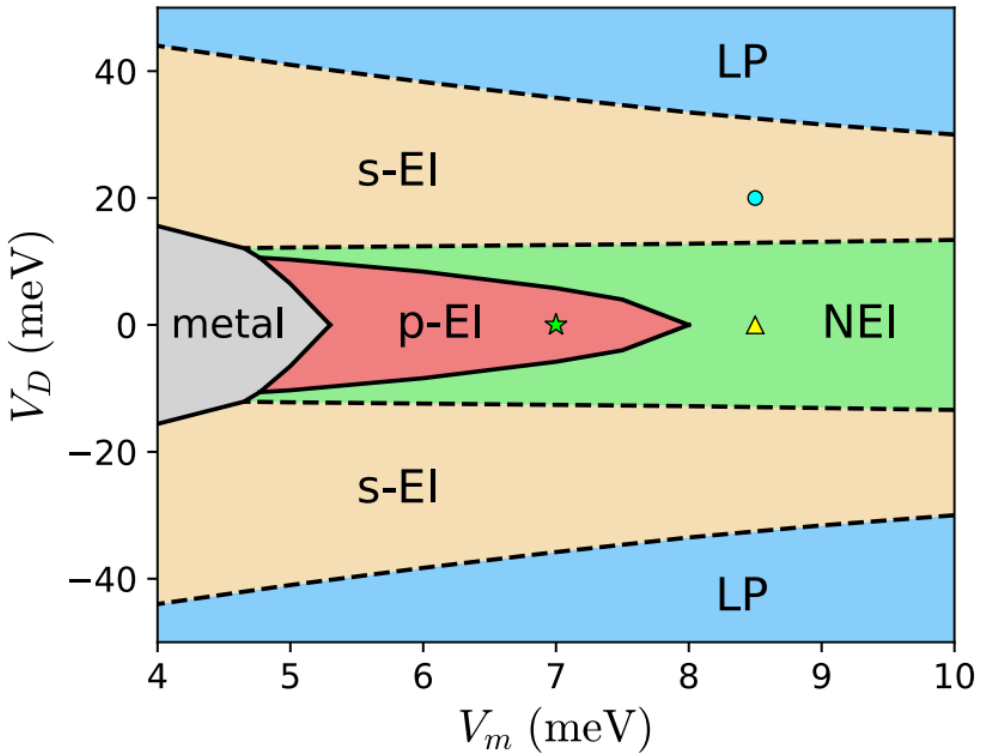


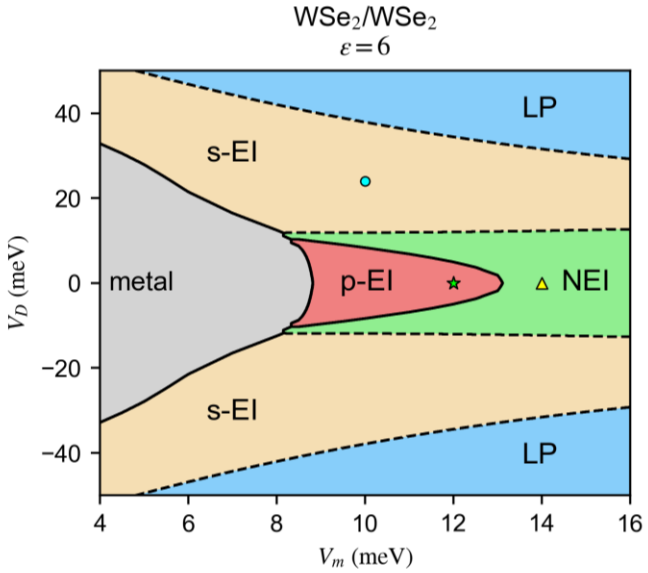


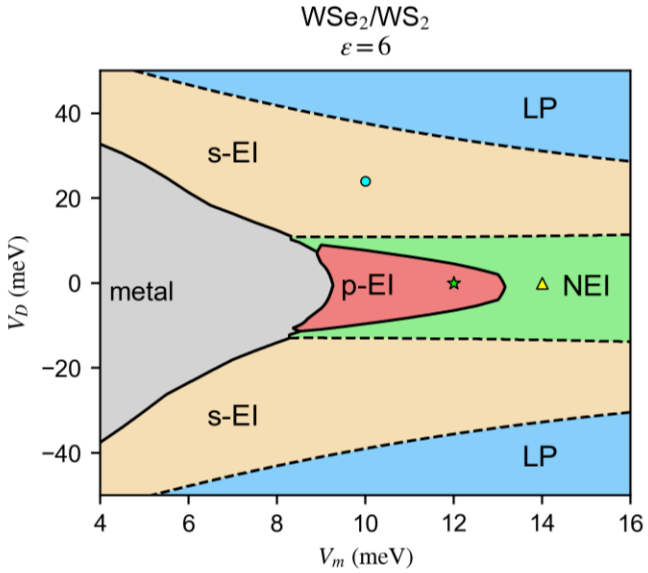
Nat. Phys. 18, 395–400 (2022); Sci. Adv. 11, eadt7789 (2025)

Phase diagram

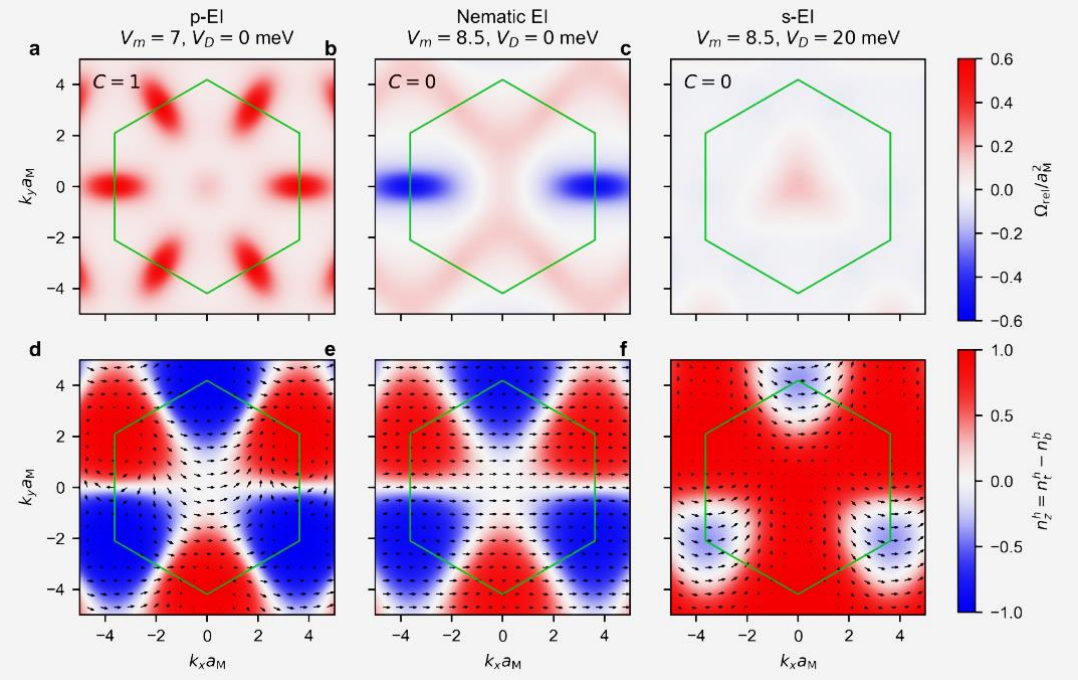
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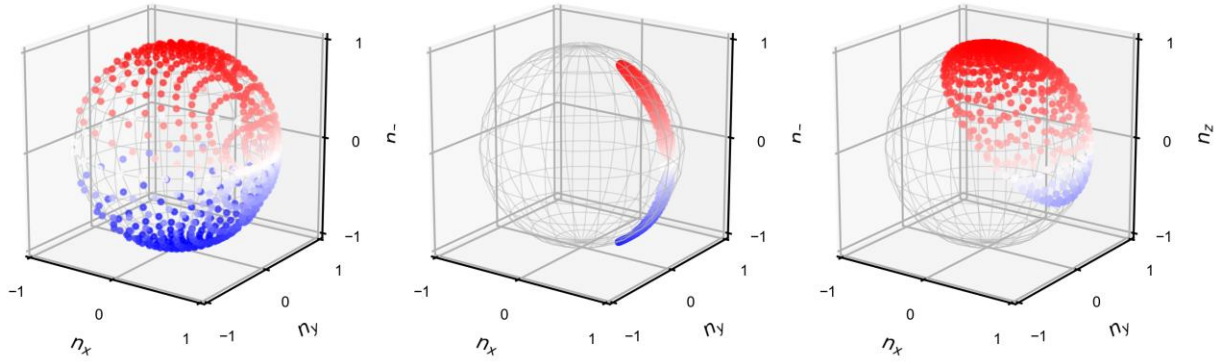


Topology calculation



Mapping the calculated moire momenta onto the pseudospin sphere

p-EI: C = 1 Nematic EI: C = 0 s-EI: C = 0

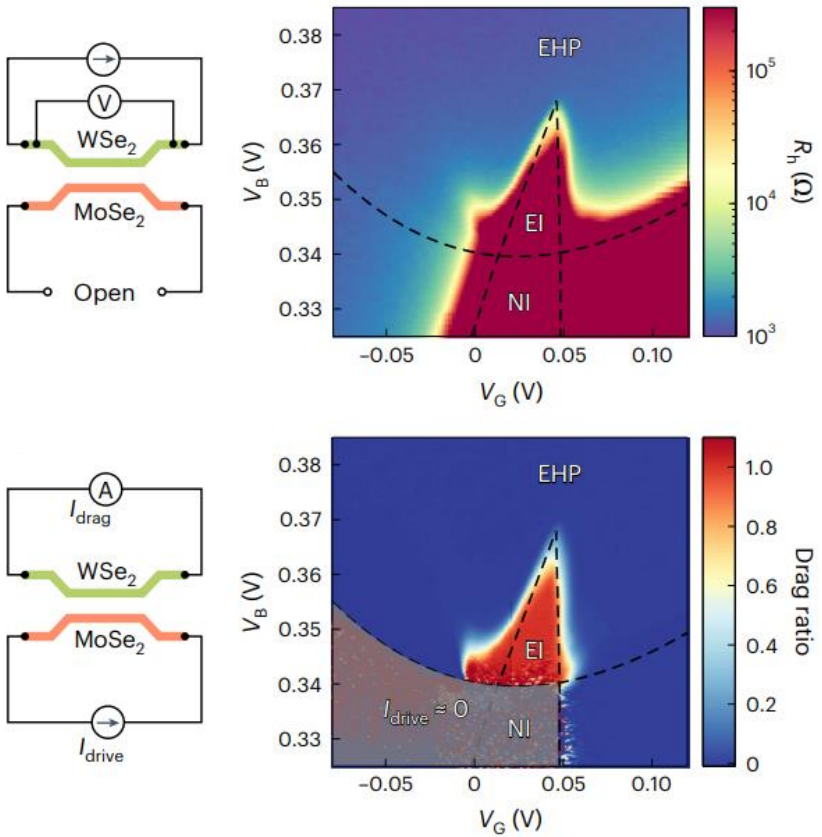


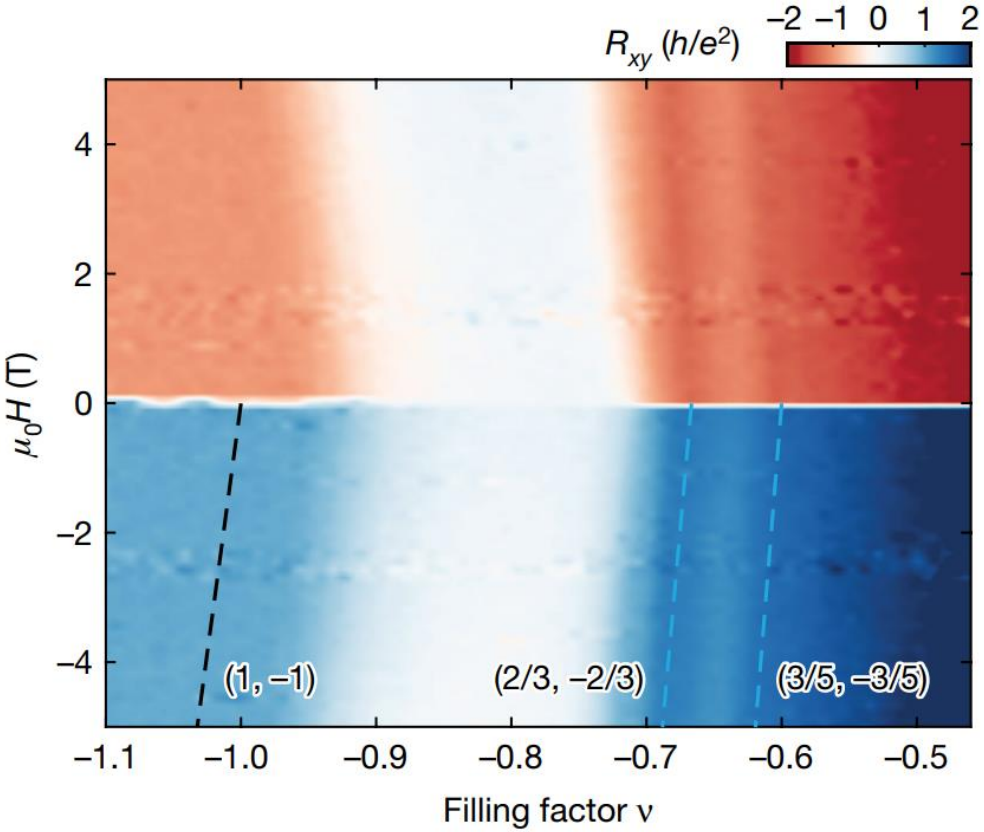
Topology nature: the electrons at κ and κ’ are fully polarized to different sublattices, and the in-plane components of sublattice pseudospins form vortices of opposite chirality around these points.

$$\Omega_{\text{coeff}} = -\frac{1}{2}\,\mathbf{n}^e \cdot \left(\partial_{k_x}\mathbf{n}^e \times \partial_{k_y}\mathbf{n}^e\right)$$

Topological exciton insulating state

Measurement: Exciton insulating state + topology
Perfect coulomb drag+ Counterflow+ Parallel flow/ Středa formula





Nat Commun 16, 178 (2025); Nature 622, 74–79 (2023)



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THANKS

Speaker: Zhongqi Xiu

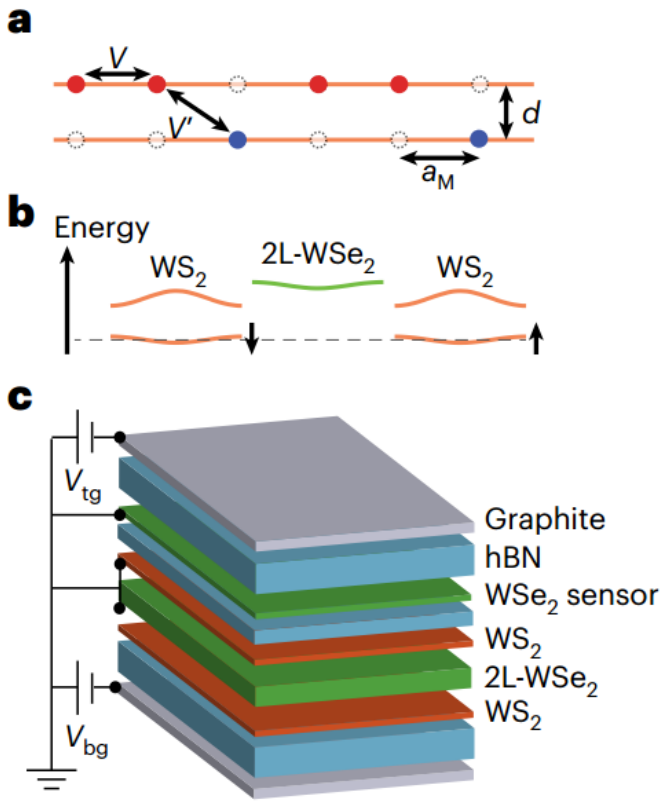
Exciton density waves in Coulomb-coupled dual moiré lattices

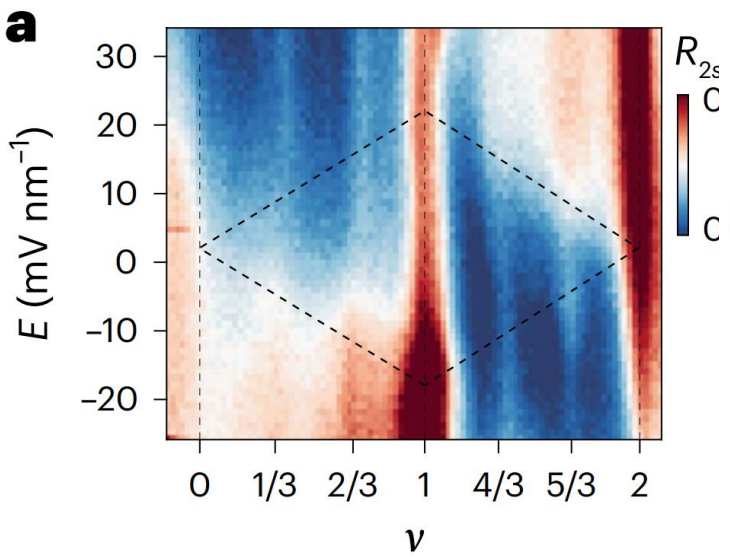
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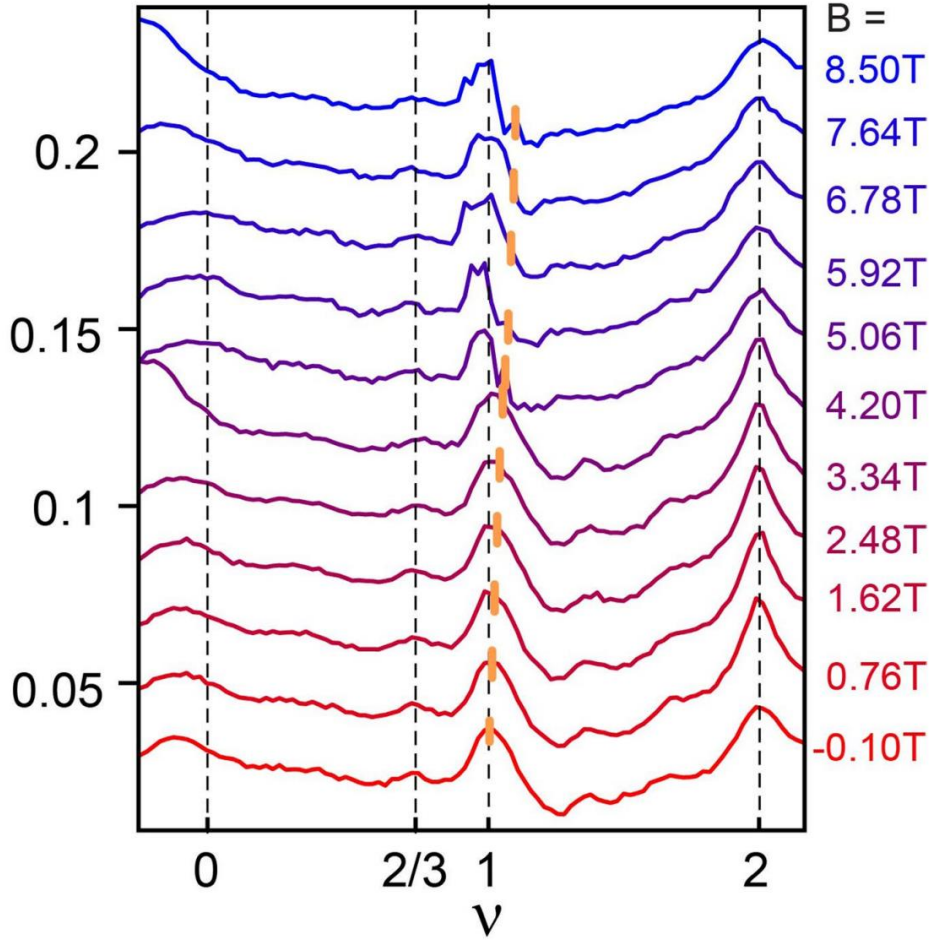
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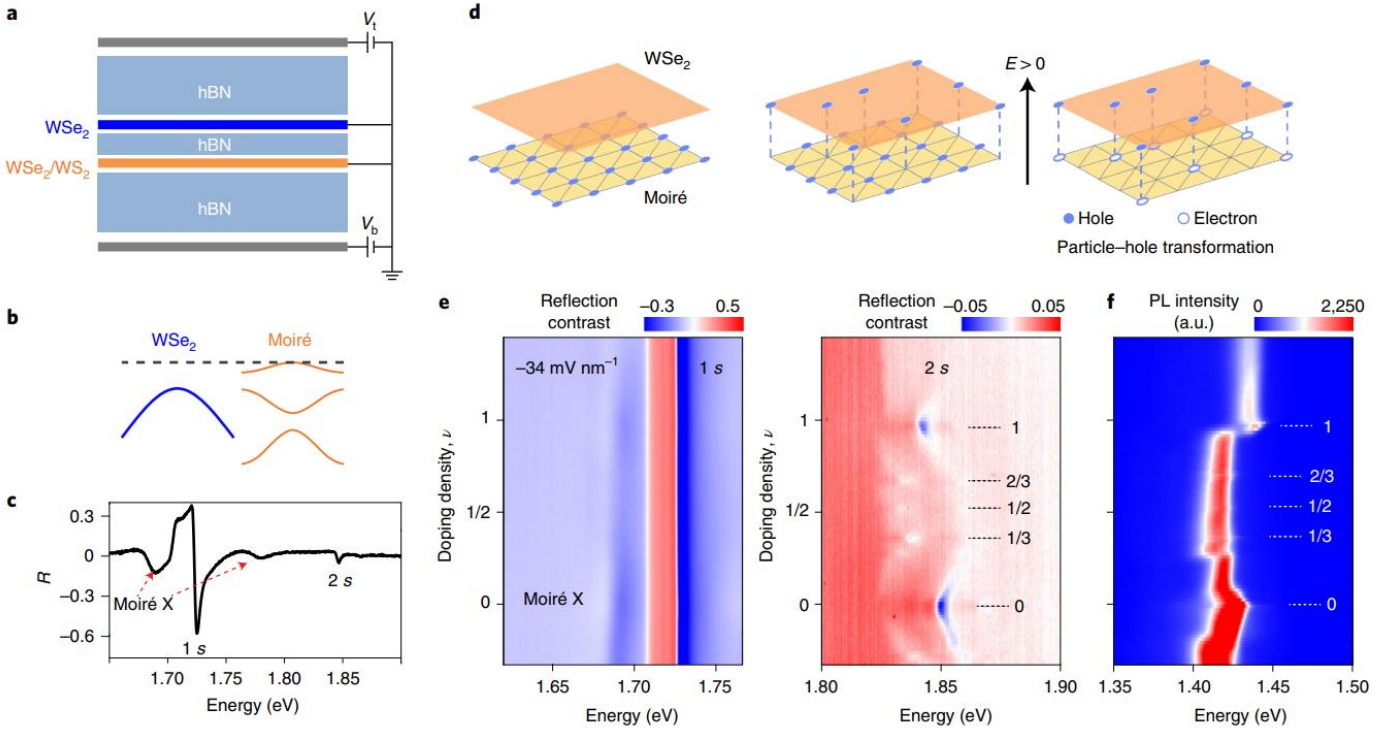


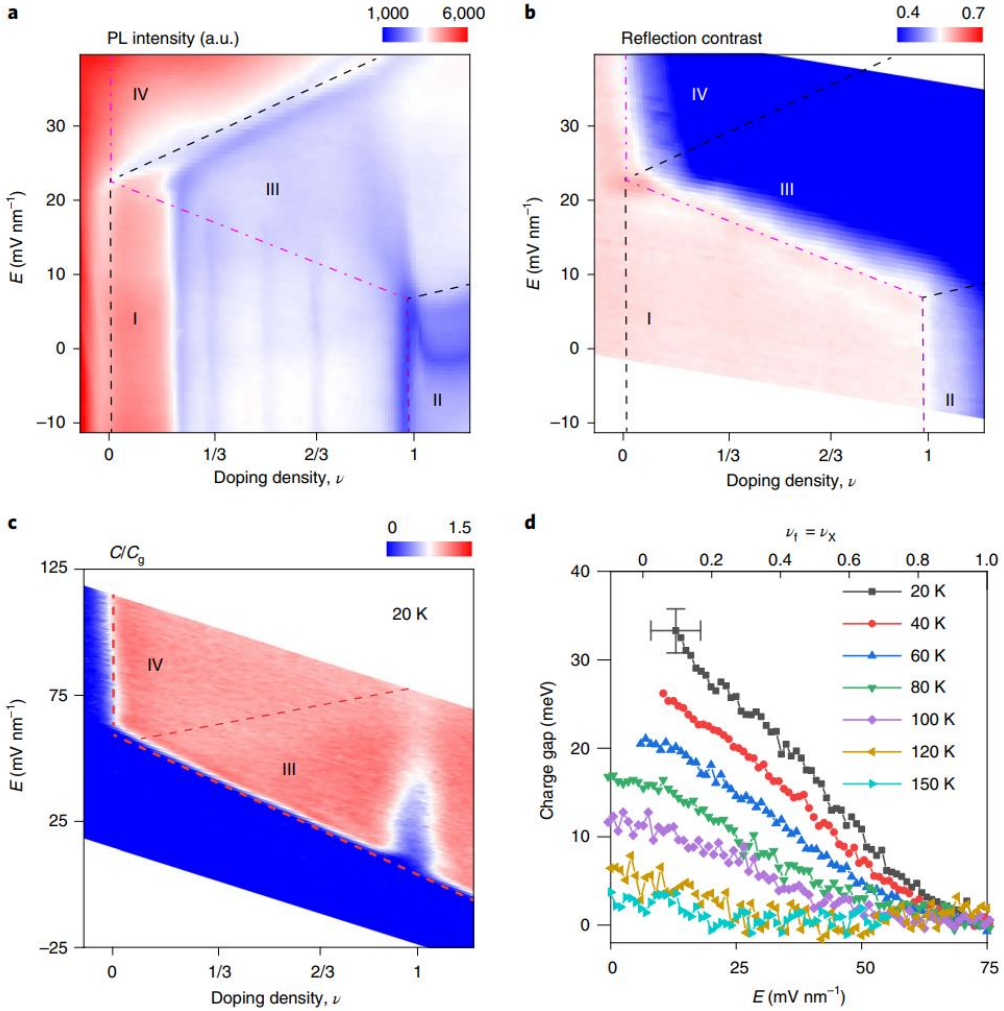
Nat. Mater. 22, 175–179 (2023)

Dipolar excitonic insulator in a moiré lattice

Jie Gu[1], Liguo Ma[1], Song Liu[2], Kenji Watanabe[3], Takashi Taniguchi[3], James C. Hone[2], Jie Shan[1,4,5] ✉ and Kin Fai Mak[1,4,5] ✉

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Nat. Phys. 18, 395–400 (2022)